Flavio A. C. Martins

Computational Engineer / CFD, Fluid-Structure Interaction & Scientific Software

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LinkedIn | GitHub | Google Scholar | Personal webpage



Profile

Computational engineer with over seven years of experience in CFD, fluid-structure interaction and scientific software for wind energy, marine and aerospace applications. I develop simulation tools from numerical method to validated result: GPU-accelerated solvers (Python, C++, CUDA), coupling with OpenFOAM, and validation against experiments.

Work experience

PhD Candidate – Computational Aerodynamics

TU Delft, Wind Energy section

Mar 2023 - Present | Delft, Netherlands

- Designed and built **OpenONDA**, an open-source Python/C++ framework that couples a GPU-accelerated vortex-particle solver with **OpenFOAM**; accepted in the *OpenFOAM Journal* (2026).
- Added **large-eddy simulation (LES)** to the particle solver and applied it to multirotor wind turbines and wind-farm momentum entrainment; published in *Wind Energy Science* (2025).
- Co-supervised an MSc thesis (GPU fast multipole method); supervise BSc project teams.

Research Engineer

RivalLab, Queen's University – project with Pratt & Whitney Canada

May 2021 - Mar 2022 | Kingston, Canada

- Calibrated a transition-sensitive **RANS** model (k-ω-γ) against experiments for transonic turbine flows; delivered an NDA-protected report.
- Designed and built towing-tank test rigs with actuation and instrumentation; acquired and processed **PIV/PTV** and force data.

Graduate Researcher (MASc) – Fluid Dynamics

Queen's University

Mar 2019 - Feb 2021 | Kingston, Canada

- Developed graph-based methods to detect coherent flow structures in sparse particle-tracking data; published in *Experiments in Fluids* (2021).
- Evaluated a marine propulsion and energy-harvesting concept with **CFD**, towing-tank tests and force measurements.

Technical skills

Programming

Python; C++; CUDA; MATLAB
Git; Linux; GPU computing

CFD & numerical methods

OpenFOAM; RANS; LES
Finite-volume and vortex-particle methods; mesh generation
Fluid-structure interaction (FSI)

Structures

FEM / FEA; Abaqus; ANSYS
Structural dynamics; vibration
CAD; SolidWorks
Load estimation; optimisation

Experiments & data

PIV / PTV; DAQ
Wind tunnels; towing tanks
Force and pressure measurements
Signal and statistical analysis
Model calibration; data assimilation
Technical reporting

Languages

English / C2

Dutch / B1

Portuguese / Native

Work experience (continued)

Research Associate – Fluid-Structure Interaction

Federal University of ABC (UFABC)

Feb 2018 - Feb 2019 | Santo André, Brazil

- Modelled **vortex-induced vibration (VIV)** of elastically mounted cylinders in 2D/3D CFD, quantifying the effects of structural damping and Reynolds number; relevant to offshore risers.
- Published in *Marine Structures* and the *International Journal of Mechanical Sciences* (2019).

Mechanical Engineering Intern

Civil Structural Engineering Ltd

Jul 2017 - Jan 2018 | São Paulo, Brazil

- Supported **civil and steel-structure projects**: client and site visits, load estimation, **CAD drawings**, technical reports and client presentations.
- Developed a **real-time data-acquisition (DAQ)** system for vibration monitoring of machine foundations, including sensors and spectral analysis.

Leadership & community

Chair, Aerospace Engineering PhD Council – TU Delft

2025-2026; member since 2023. Redesigned the annual PhD symposium (nearly 200 attendees), reduced its budget from EUR 20,000 to EUR 5,000 and secured industry sponsorship.

Member, University PhD Council – TU Delft

2026. University-wide events: planning, budgets and stakeholder coordination.

Dutch OpenFOAM user community

2023-2026. TU Delft representative; co-organised the national user meetings.

Teaching

Teaching assistant for experimental fluid-dynamics labs, Queen's University (2019-2021); MATLAB and Abaqus workshops for engineering students (2017-2018).

Science outreach

OffshoreWind4Kids team leader (2024-2026); developed *Wind for Kids* teaching materials for primary schools (2024-2025).

Education

PhD, Aerospace Engineering

TU Delft
2023 - Present

MASc, Mechanical & Materials Engineering

Queen's University, Canada
2019 - 2021

BSc, Aerospace Engineering

UFABC, Brazil
2016 - 2018
Valedictorian; Institute of Engineering Award

BSc, Science & Technology

UFABC, Brazil
2012 - 2016

Publications

7 journal articles and 9 conference papers in CFD, FSI, wind energy and structural optimisation. Full list on Google Scholar.

Hobbies

Photography

Cycling and running

Painting

Music production